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Experiment 7:

AIM: Implementation of KNN using sklearn

PROGRAM:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report
iris = load_iris()
data = iris.data
target = iris.target
X_train, X_test, y_train, y_test = train_test_split(data, target, test_size=0.3)
clf = KNeighborsClassifier(n_neighbors=5)
clf.fit(X_train, y_train)
predict = clf.predict(X_test)
print(f"Scikit-learn KNN classifier accuracy: {accuracy_score(y_test, predict)}")
print(classification_report(y_test, predict))
```

Output:

```
Scikit-learn KNN classifier accuracy: 0.9555555555555556
              precision    recall  f1-score   support
0                 1.00      1.00      1.00        14
1                 0.92      0.92      0.92        13
2                 0.94      0.94      0.94        18
accuracy                              0.96        45
macro avg         0.96      0.96      0.96        45
weighted avg      0.96      0.96      0.96        45
```